

Radiomics-Enhanced Diabetic Retinopathy Detection from Retinal Fundus Images

Ansh Bajpai

*Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Amritapuri, India
AM.SC.U4AIE23108*

Nishant Shah

*Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Amritapuri, India
AM.SC.U4AIE23166*

Abstract—The most common causes of blindness that doctors can prevent globally are Diabetic Retinopathy (DR). Therefore, it is essential to accurately assess the degree of DR. Although AI-based computer vision systems have greatly improved automated detection of DR using Deep Learning, many current methods utilize only one data extraction path to extract features and therefore fail to capture the complex relationships between the overall structure of the retina; the detailed characteristics of the lesions; and the subtle changes in the small blood vessels. When trying to distinguish between grades of DR that are visually similar, these factors are crucial. The potential to improve upon the current limitations of many existing approaches will be exacerbated by issues such as large differences in sample size among classes; large variations within each grade-level category; and limited utilization of additional information available, including radiomic information about the texture and/or intensity of structures; and finer-resolution information about the structural characteristics of the vessels.

Therefore, this paper proposes a new, general-purpose multi-stream fusion framework to fully integrate and combine all of the various retinal cues. In addition to semantic features derived from the state-of-the-art backbone network architectures, our approach includes the inclusion of radiomics-derived texture and intensity descriptors; vessel maps that highlight structural pathological changes; and lesion-focused patches that were obtained through targeted region analysis. To fuse these disparate sources of information into a single embedding space where the model can learn meaningful associations between different types of feature information, a cross-modal alignment/fusion module is specifically designed for this purpose. Additionally, to further improve the robustness of the proposed system across all DR severity levels, we also include two new components: a balanced sampling strategy and a multi-loss optimization scheme.

When we tested the method on the DDR multiclass dataset, it reached an accuracy of 90.42 %, a QWK of 94.17 % and a weighted average F1-score of 90.01 %. Those values surpass both standard baseline models plus earlier published results on the same benchmark. The gains show that the combined use of structural information, radiomic features and lesion-focused indicators produces a representation of retinal pathology that is both more discriminative and closer to clinical needs.

In summary, our work establishes a new benchmark for multi-class DR severity grading and introduces a generalizable, lesion-aware framework that can support future medical imaging systems that requires fine-grained and clinically meaningful decision making.

Index Terms—Diabetic Retinopathy, multi-class severity grading, deep learning, multi-stream fusion, radiomics features, vessel maps, lesion-aware features.

I. INTRODUCTION

The growing number of people who suffer from Diabetic Retinopathy (DR) around the world is now causing problems for eye care professionals in terms of time to assess patients and cost. In addition, there is an increasing urgency to develop an early DR diagnosis system. Therefore, deep learning-based systems have received considerable interest lately due to their ability to quickly evaluate thousands of images with each new input and learn through practice. However, most of these systems can only review images as a whole rather than at the individual lesion or disease level which clinicians rely on when assessing the degree of DR. Due to this limitation, whole image based diagnostic systems can easily misclassify adjacent levels of retinopathy and generate descriptions that are too ambiguous to be relied upon by practicing clinicians.

Progress has been made toward image classification using Convolutional Neural Networks (CNN) as well as attention-based models; however, current Deep Learning (DL) based DR classifiers are essentially "black boxes" with no visibility into their internal workings. While Grad-CAM based heatmaps can highlight general regions of interest for a given image, they do not provide sufficient detail to identify and describe individual diseased locations within an image. In addition to lack of clarity regarding lesion specific details, CNN feature extraction is also limited by the fact that features extracted from images are often too general to distinguish between mild, moderate, and severe DR, and therefore require additional processing techniques. Therefore, we believe it is important to develop methods for extracting, displaying, and analyzing lesion information in a format that meets clinical requirements and operates independently.

Although previous research has addressed several of these issues in part, current solutions are not sufficient to fully meet the needs of clinicians, researchers or computer scientists. Current CNN-based models achieve significant advances in global lesion classification, however they provide limited insight to how they have identified specific lesions (i.e., lesion-level interpretation). While numerous Radiomics-based studies have been successful at extracting hand-crafted radiomic features from imaging data that can be used to predict patient outcomes, most require manual segmentation

of lesions; this is impractical for large-scale clinical use. The ability of multi-modal fusion approaches to combine clinical metadata with other data types (e.g., imaging) does not address the spatial relationship between lesions. More sophisticated models (e.g., FA+KC-Net, Attention-Driven Deep Learning Architectures) enhance the ability to identify where within an image a model has localized a feature, however both continue to treat lesions as continuous (or "diffuse") areas of disease rather than discrete pathological units. To date, there exists no unifying framework that combines: (1) CAM-driven discovery of lesions from images, (2) automated extraction of radiomics from images, (3) deep feature representations of the images and (4) graph-based relational reasoning to understand the relationships between the lesions.

In terms of the challenge of fundus image analysis, we have identified several obstacles. To address these challenges, we used Lesion-Level Patch Intelligence Framework. This framework will be developed based on a hierarchical approach to processing fundus images. Fundus images will be broken into smaller and clinically relevant regions (patches) and then processed at a region-specific level in a sequential manner to accumulate information. The sequence begins with Clinical Algorithm Model (CAM)-guided maps of the fundus to identify isolated lesion patches, which were determined using Otsu-based thresholding. This method allows for scaling up lesion detection without requiring manual labeling. Each patch will contain a more comprehensive representation. Specifically, it combines the rich semantic detail from ConvNeXt with basic radiomics attributes. Basic radiomics include statistics (first order), GLCM, GLRLM, and GLSZM. Thus, it provides a combination of a global perspective as well as local texture cues. Additionally, we will add spatial information to every patch including bounding boxes and CAM strengths. By doing so, we enable the model to reason about the location of lesions and their relevance. Once all patches are generated, they are merged into one lesion graph with nodes representing individual lesions and edges representing their relationships in space. A graph neural network will then process the relationships between lesions and generate a more complete view of disease progression. Finally, we will combine the lesion embedding representations with the radiomics data and graph context to make a DR severity assessment that is clinically relevant and logically consistent.

This effort brings new, fully automated setup for grading DR from lesions. It centers everything on those lesions. The result boosts how accurate predictions get, along with clearer explanations. We connect deep learning more closely to how clinicians look at lesions. The system edges toward actual use in clinics. It provides a clearer, more useful way to check DR.

II. LITERATURE REVIEW

Automated diabetic retinopathy (DR) detection has gained significant traction as deep learning models continue to advance, offering scalable diagnostic support for overwhelmed clinical systems. Early research established the feasibility of CNN-based DR grading, with lightweight architectures such

as MobileNet and MobileNetV2 demonstrating competitive performance on imbalanced DDR datasets for both binary and five-class classification tasks [4]. More sophisticated approaches, such as the FA+KC-Net fine-grained attention and knowledge-based collaborative framework, addressed subtle lesion identification by combining patch-level attention with medically informed lesion detectors, achieving state-of-the-art results on DDR, Messidor, APTOS, and EyePACS [1]. Transfer learning has also been widely explored; ResNet50-based models trained on large-scale datasets like EyePACS have achieved validation accuracies above 95%, demonstrating the strong capability of deep residual networks in fundus-based DR classification [5]. Multimodal systems have further extended DR analysis by integrating clinical biomarkers such as HbA1c with CNN features; ensemble models combining EfficientNet and GoogLeNet have reported accuracies as high as 98.6%, highlighting the predictive value of combining retinal imaging with systemic diabetic indicators [6]. Other works explore alternative optimization strategies, including neuroevolution and hybrid ensembles, to improve model robustness on small datasets, offering viable solutions for resource-limited settings but often restricted to only 2–3 DR classes [3]. Ensemble-based lightweight CNN classifiers have also been proposed to improve multi-class grading using one-vs-one strategies on datasets such as APTOS, IDRiD, and DDR, though these methods typically combine predictions rather than providing lesion-level interpretability [2].

Although improvements have been made with CNN's, there are still many limitations to current models. Many CNN architectures are built to learn global (holistic) image characteristics as opposed to learning about specific lesion locations within images. The lack of a method to reason at the lesion level has led to poor classification of mild/moderate DR patients who typically have very small and/or subtle lesions that are difficult to detect by machine vision alone. While CAM based visualization provides a general "heat map" of activity across an image it does not clearly define specific areas of interest (lesions). Thus the results are often non-interpretable clinically. Radiomic analysis is often reliant upon manual segmentation of lesions, thereby limiting its feasibility for automated screening applications. Additionally, since many multimodal systems utilize HbA1c levels, and other forms of clinical data, they are limited in their deployment to regions without access to those types of data. Finally, even though systems such as FA+KC-Net use pre-trained lesion detection models, and ensemble systems can significantly enhance predictive accuracy, neither of these approaches provide meaningful insights into how individual lesions interact with each other, nor do they help to explain the underlying mechanisms of disease progression.

All of these limitations will be addressed through the development of an end-to-end fully automated Lesion Level Patch Intelligence Framework. This framework is designed to produce lesion-level patches at their most granular state using Class Activation Map (CAM) based segmentation; therefore, there is no need for any form of manual annotation of lesion patches or use of external lesion detection software. The lesion

patches will be represented by a dual feature embedding that includes both deep ConvNeXt-based semantic features and hand-crafted radiomics descriptors that allow for quantification of both visual and micro-textural characteristics of DR pathology. Also, the inclusion of spatial information as part of the lesion graph that has undergone Graph Neural Network (GNN) processing allows for the model to describe how lesions interact with one another and the context in which they exist that does not occur with contemporary CNN and ensemble architectures. The integrated framework of this paper creates a connection between the clinical interpretation of lesions and the image-level deep learning, and produces predictive results that are both more accurate than those produced by existing approaches and inherently interpretable and consistent with the clinical ophthalmic diagnostic process.

III. METHODOLOGY

We present a grading system for Diabetic Retinopathy that takes multiple kinds of data into account. It uses convolutional neural backbones, handcrafted radiomics, a fusion mechanism guided by radiomics and modules that reason at the level of individual lesions. The system is tuned to match the ordinal scale of DR severity - maximizing the Quadratic Weighted Kappa and it moves past simple image level labels to supply clear lesion based explanations.

A. Dataset and Preprocessing

Dataset Description

We utilize two publicly available retinal fundus datasets for diabetic retinopathy (DR) severity classification: EyePACS and DDR.

EyePACS Dataset: This large-scale dataset comprises approximately 88,702 retinal fundus images collected from diverse clinical sources under varying imaging conditions. Each image is labeled according to the standard 5-class DR grading scale:

- 0 — No DR
- 1 — Mild
- 2 — Moderate
- 3 — Severe
- 4 — Proliferative DR

EyePACS was used for pretraining a ResNet50 backbone to establish a robust retinal feature prior.

DDR Dataset: The Joint Shouthern International Eye Center supplied the DDR set, which holds 13,673 sharp fundus pictures. The set uses the same five grade scale for diabetic retinopathy that EyePacs uses - most pictures fall into the two lowest grades - no retinopathy or mild retinopathy - just as in real screening programmes. DDR adds value to EyePacs because every picture keeps the same sharpness and lesions stand out clearly - researchers can pull radiomic features and study each lesion in detail.

Every picture was brought down to 512×512 pixels. First the black border was cut away. Next the grey world method balanced the light across the image. Finally CLAHE raised the contrast on the brightness channel so that microaneurysms,

exudates plus blood-vessel patterns became easier to see. For lesion-level modeling, CAM-guided segmentation was used to extract focused patches of size 224×224 .

DDR fundus images with corresponding DR grades serve as the primary dataset.

Preprocessing steps

- 1) Image integrity inspection and removal of corrupted files.
- 2) Border cropping and resizing to 512×512 .
- 3) Gray-world normalization to correct illumination.
- 4) CLAHE applied on the luminance channel to highlight vasculature and lesions.

These procedures create uniform, clinically meaningful representations suitable for both CNN feature extraction and radiomics computation.

B. Backbone Pretraining and Embedding Extraction (Group 1)

We train three complementary visual backbones to extract global retinal representations:

- EfficientNet-B3 $\rightarrow$ 1536-D embedding (QWK $\approx$ 0.838)
- ConvNeXt-Small $\rightarrow$ 768-D embedding (QWK $\approx$ 0.908)
- Swin-Tiny $\rightarrow$ 768-D embedding (QWK $\approx$ 0.878)

Each architecture captures different retinal cues: EfficientNet emphasizes fine lesion detail, ConvNeXt captures global structural context, and Swin encodes long-range dependencies. Checkpoints and per-backbone Grad-CAM heatmaps are saved as primary artifacts for downstream lesion localization.

C. Radiomics Extraction (Group 2)

Handcrafted radiomics provides clinically interpretable descriptors that complement deep embeddings. The radiomics pipeline produces a fixed-length vector per image/patch including:

- First-order intensity descriptors
- Texture signatures (GLCM, GLRLM, GLSZM)
- Vessel geometry statistics and lesion morphology metrics

Earlier versions used a larger feature set; the pipeline is configurable. The reported Group 3 fusion used a 107-D radiomics vector as input. All radiomics features are standardized (zero mean, unit variance).

D. Multimodal Fusion Architecture (Group 3 — Core Contribution)

Objective. Build a radiomics-guided fusion head that combines multiple pretrained visual backbones with clinical measurements to yield a clinically consistent severity representation.

Per-modality projection:

1536 $\rightarrow$ 512	EfficientNet-B3
768 $\rightarrow$ 512	ConvNeXt-Small
768 $\rightarrow$ 512	Swin-Tiny
107 $\rightarrow$ 128 $\rightarrow$ 512	Radiomics MLP

Radiomics-guided cross-attention. Radiomics acts as a query attending to the visual tokens (keys/values). An 8-head multi-head cross-attention mechanism (embed_dim = 512) aligns visual modalities to clinical priors, allowing the system to dynamically select which backbone(s) to weight for each case.

Prediction heads. The fused token is refined by MLP layers and passed to two heads:

- 1) Softmax classifier (5 classes)
- 2) CORAL ordinal head (cumulative logistic thresholds) to encode label ordering

Training objective.

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{CORAL}, \quad \alpha = 0.2$$

Optimization used AdamW (lr = 3×10^{-4} , wd = 10^{-4}), CosineAnnealingLR, 20 epochs. Reported performance: QWK ≈ 0.9417 , Accuracy ≈ 0.9042 .

Group 3 is the main methodological advance: by forcing visual embeddings to align with clinically interpretable radiomics, the fusion head produces more clinically consistent and ordinally correct predictions than simple concatenation.

E. Lesion-Level Pipeline and Node Representations (Group 4 — Lesion Patch Extraction & Encoding)

We did not stop at image-level fusion. To add lesion-level interpretability and relational reasoning, Group 4 converts Grad-CAM localization into lesion nodes that feed a graph pipeline.

Steps performed in Group 4 (single integrated workflow):

- 1) **Grad-CAM generation:** From the pretrained backbones (Group 1), per-image Grad-CAM heatmaps were produced (we used ResNet-50 for Grad-CAM generation).
- 2) **Masking and morphology:** Each Grad-CAM heatmap is normalized, thresholded using Otsu, and cleaned with morphological opening/closing to remove noise and consolidate lesion blobs.
- 3) **Connected components and bounding boxes:** Connected components are extracted and filtered using a minimum area threshold (300–350 px). Bounding boxes (x_1, y_1, x_2, y_2) are computed for each valid lesion.
- 4) **Patch cropping and metadata:** Lesion patches (128×128 or 224×224) are cropped and recorded in `patch_metadata.csv` containing: `class_id`, `image_id`, `patch_id`, `patch_path`, `x1`, `y1`, `x2`, `y2`, `cam_score`.
- 5) **Patch embeddings:** Each lesion patch is encoded using a ConvNeXt-Small model (separate from Group 3) to produce a 768-D embedding.
- 6) **Patch radiomics:** PyRadiomics extracts first-order, GLCM, GLRLM, and GLSZM descriptors (74-D in our implementation).
- 7) **Metadata:** Bounding box coordinates are normalized and the mean CAM intensity stored as `cam_score`.

Final node representation.

Patch embedding (768-D) + Radiomics (74-D)
+ Metadata (5-D) $\Rightarrow$ **847-D** node vector

Group-4 artifacts and scale.

- Detected patches: 23,080
- Embeddings: (23080, 768), Radiomics: (23080, 74), Coords: (23080, 4)
- Unique images covered: 12,460

Why Group 4 matters. Group 3 yields high QWK at image-level but cannot explain which lesions drive predictions. Group 4 introduces explicit lesion nodes that:

- localize visual evidence,
- attach interpretable radiomics descriptors,
- encode spatial context with bounding boxes and CAM scores,
- enable relational modeling of lesion interactions in Group 5.

F. Lesion Graph Construction and GNN Reasoning (Group 5 — In Progress)

Graph construction strategy.

- One graph per image (nodes = lesion patches)
- Node features: 847-D vectors
- Edges: k -nearest neighbors using normalized lesion coordinates

GNN outputs.

- Image-level DR grade via global pooling
- Lesion-level importance via attention or node relevance scores

Scientific rationale. Lesion co-occurrence and spatial patterns (e.g., hemorrhage clusters) encode disease progression signals not captured by global image embeddings. GNNs enable reasoning over these interactions.

G. Implementation and Reproducibility

Experiments used PyTorch, timm backbones, PyRadiomics, and PyTorch-Geometric. Checkpoints include `best_by_qwk` and `last_epoch`. All metadata and intermediate files are reproducible at the specified paths.

Takeaway. We did not stop at a high QWK classifier. Group 4 transforms the model into a lesion-aware reasoning system capable of explaining its decisions and supporting clinically actionable insights.

IV. PERFORMANCE EVALUATION

A. Experimental Setup

All experiments were conducted using the DDR dataset for 5-class Diabetic Retinopathy (DR) grading. The entire pipeline was implemented in Python using PyTorch 2.1, CUDA acceleration, and executed on an NVIDIA Tesla P100 GPU (Kaggle environment), ensuring reproducibility and stable runtime.

The proposed model follows a unified end-to-end architecture consisting of: (i) three backbone feature extractors (EfficientNet-B3, ConvNeXt-Small, Swin-Tiny), (ii) a radiomics encoder that embeds vessel density, branching patterns, entropy, and lesion texture features, (iii) projection to a shared 512-dimensional latent space, (iv) cross-attention

fusion to integrate multimodal evidence, (v) classification & ordinal regression heads.

Training was performed using the AdamW optimizer with cosine annealing, batch size of 16, and an initial learning rate of 1×10^{-4} . Weighted cross-entropy was used to address class imbalance. To enforce ordinal consistency in DR grades (0–4), CORAL loss was jointly optimized with the CE loss.

Grad-CAM visualizations were generated post-training to ensure interpretability, and all lesion patches were processed using our Grad-CAM-guided cropping pipeline combined with radiomics extraction.

B. Evaluation Metrics Used

To provide a comprehensive assessment of DR grading performance, several metrics were used.

1) Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

2) *Weighted F1-Score*: Weighted F1 accounts for class imbalance by weighting each class by its support:

$$\text{Weighted-F1} = \sum_{c=1}^C w_c \cdot \frac{2 \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

where $w_c = \frac{N_c}{\sum_c N_c}$.

3) *Quadratic Weighted Kappa (QWK)*: QWK measures agreement between predictions and true labels while penalizing ordinal distance.

$$\kappa = 1 - \frac{\sum_{i,j} W_{ij} O_{ij}}{\sum_{i,j} W_{ij} E_{ij}}$$

where O is the observed matrix, E is the expected matrix, and:

$$W_{ij} = \frac{(i - j)^2}{(C - 1)^2}$$

4) Cross-Entropy Loss:

$$\mathcal{L}_{CE} = - \sum_{c=1}^C y_c \log(\hat{y}_c)$$

5) *CORAL Ordinal Loss*: To enforce ordered severity levels:

$$\mathcal{L}_{CORAL} = - \sum_{k=1}^{K-1} \left[y^{(k)} \log \hat{p}^{(k)} + (1 - y^{(k)}) \log(1 - \hat{p}^{(k)}) \right]$$

C. Quantitative Results

The proposed model achieves strong performance across all evaluation metrics. Our best model reached:

- **Accuracy:** 90.42%
- **QWK:** 94.17 %
- **Weighted F1-Score:** 90.01%
- **Loss components:** CE + $0.2 \times$ CORAL

The high QWK score demonstrates that the model effectively maintains ordinal consistency between DR grades. Confusion matrices and ROC/F1 curves (shown in subsequent

figures) further validate the improved classification of Grades 1–3, which are typically challenging due to subtle inter-class variations.

The fusion of image embeddings, lesion-level radiomics, and cross-attention clearly increases robustness, as reflected by improved mid-grade discrimination compared with baseline CNN architectures and prior literature models.

TABLE I
CLASSIFICATION REPORT OF GROUP 3 FUSION MODEL

Class	Precision	Recall	F1-Score	Support
0	0.9289	0.9734	0.9566	940
1	0.5000	0.4211	0.4571	95
2	0.9155	0.9033	0.9094	672
3	0.6000	0.3429	0.4364	35
4	0.9542	0.9124	0.9328	137
Accuracy	0.9042 (1879 images)			
Macro Avg	0.7797	0.7106	0.7373	1879
Weighted Avg	0.8982	0.9042	0.9001	1879

TABLE II
RESULTS OF DIFFERENT BACKBONE MODELS AND THE PROPOSED MULTIMODAL DR MODEL

Model	Validation Accuracy	QWK Score
EfficientNet-B3	0.8544	0.8779
ConvNeXt-Small	0.8760	0.8760
Swin-Tiny	0.8544	0.8779
Multimodal DR (Proposed)	0.9042	0.9417

V. CONCLUSION

Diabetic Retinopathy (DR) severity grading is a complex task because of subtle lesion patterns, diverse retinal textures, and strong ordinal relationships across disease stages. In this work, we developed a multimodal DR analysis framework that progresses beyond conventional CNN based classification and towards a clinically aligned, interpretable, and progression aware prediction pipeline.

Our approach combines three complementary visual backbones, EfficientNet B3, ConvNeXt Small, and Swin Tiny, together with handcrafted radiomics extracted from fundus images. These features are merged using a radiomics guided cross attention fusion mechanism along with a CORAL based ordinal prediction head. This architecture enforces severity aware learning and produces representations that reflect clinical DR progression. Our strategy delivered substantial performance gains and we achieved a QWK score of approximately 0.94, which exceeds standalone CNN models and recent DR benchmarks.

The key contribution of this work is the demonstration that radiomics can serve as a semantic controller for attention-based fusion. This guidance allows the model to focus on medically meaningful retinal features which affect DR severity. In contrast to previous works based on simple concatenation, of features or independent multimodal streams, our design yields severity-ordered embeddings better aligned with Expert diagnostic reasoning and increased reliability across all DR categories.

TABLE III
LIMITATIONS OF CONVENTIONAL DR APPROACHES AND THE CORRESPONDING ADVANCEMENTS INTRODUCED BY OUR PROPOSED FRAMEWORK

Pipeline Weakness	Cause	Impact	Solution	What Our Component Added
Missed tiny lesions (MA, exudates)	ConvNeXt/Swin lack fine-grain focus	Confusion in Grade-1 & Grade-2	EfficientNet-B3	High-resolution local lesion detection
Poor global retinal context	EfficientNet sees patches, not full structure	Under-grading Moderate/Severe DR	ConvNeXt-Small	Large receptive fields + contextual reasoning
Weak spatial relation modelling	EfficientNet & ConvNeXt are local filters	Moderate $\leftrightarrow$ Severe confusion	Swin Transformer	Attention captures inter-lesion & vessel relations
CNNs detect texture but don't quantify it	All CNN backbones	Unstable mid-grade separation	Radiomics	Numeric texture & intensity descriptors $\rightarrow$ measurable severity
No explicit vascular health tracking	All CNNs	Miss key DR biomarkers	Vessel Metrics	Vessel density, branching, dropout $\rightarrow$ clinical vascular indicators
Single-model bias	Any backbone alone	Overfitting to one visual bias	Three-Backbone Fusion	Complementary views cancel blind spots
Ensembles combine outputs, not understanding	Softmax/Logit averaging	No severity order $\rightarrow$ lower QWK	Embedding Fusion (MLP)	Learns how much to trust each backbone feature
Classes treated as independent labels	Standard CE loss	Random jumps in prediction	Ordinal-Aware Fusion	Severity-aware decision boundaries
Mild DR looks like Normal retina	Early DR subtle changes	Worst recall for Class-1	Radiomics + Swin + Fusion Head	Texture + vessel cues + spatial context resolve ambiguity

TABLE IV
COMPARISON OF OUR PROPOSED DR GRADING FRAMEWORK WITH EXISTING STATE-OF-THE-ART MODELS ACROSS ARCHITECTURE, PERFORMANCE, AND CLINICAL RELEVANCE DIMENSIONS.

Metric / Component	Our Model	FA-Net (FA-Net = Fine-grained Attention Network) [1]	FA+KC-Net(R2) (KC-Net = Knowledge-based Collaborative Network) [1]	Lightweight Transfer Learning Ensemble (2025) [2]
Classification Type	5-Class DR (Ordinal + Classification)	5-Class	5-Class	5-Class (converted to 10 binary classifiers)
Datasets Used	DDR, EyePACS	DDR, APTOS, Messidor, EyePACS	Same	APTOS-2019, IDRiD, Messidor-2, DDR
Accuracy	90.42%	82.10%	84.49%	92% (binary models individually), ensemble approx 89–91%
QWK (weighted)	94.17%	82.68%	86.17%	Not reported (binary metrics only)
Backbone	EfficientNet-B3, ConvNeXt-Small, Swin	ResNet50	ResNet50 + YOLOv5 lesion detector	MobileNet, InceptionV3, DenseNet121
Feature Fusion	Cross-Attention Fusion (learns which backbone to trust)	Simple attention	Rule-based feature + lesion logic	Majority voting
Multimodality	Yes (Image + Radiomics)	No	No	No
Explainability	Grad-CAM + patch-level radiomics	None explicitly	Lesion rules (handcrafted)	None beyond saliency maps
Ordinal Modeling	CORAL Loss (ordered thresholds)	No	No	No
Handling Confusing Grades	Radiomics + cross-attention improves Mild/Moderate separation	Poor Mild/Normal separation	KC-Net fails for DR4	Paper highlights difficult class overlaps
Training Pipeline	End-to-end unified	Two-stage	Two modules + rules	10 separate binary models (OVO)
Inference Stability	High (single model, no rules)	Moderate	Fails for DR4	Unstable (10 models + voting)
Clinical Relevance	Vessel metrics + texture severity	Limited	Moderate (lesion rules)	None

Although the current system achieves state of the art performance at the image level, several advanced components of the extended workflow are still under development. Group 4 introduces lesion-specific patch extraction and embedding by using Grad CAM-based localization. Similarly, group 5 utilizes GNN which determine the relationship among lesions. Group 7 aims to improve concept-based interpretability via Group 8 and TCAV seek to generate one DR knowledge graph to encourages the provision of clear, rule-based diagnostic explanations. These additions are expected to transform the framework from a high performing classifier into a clinically grounded reasoning system that links predictions to lesion behaviour and anatomical context.

Although the model performed well with the data it was

trained on, it is still dependent upon high-resolution images, and also does not currently take advantage of longitudinal patient history. The future development of this model will involve addressing these issues by incorporating lesion progression signals into the model, including any relevant clinical metadata when available, and developing expanded graph reasoning capabilities that allow for the model to be justified based on patient-specific information.

Ultimately, this research establishes a strong foundational architecture for the interpretability of DR severity classification using integrated deep feature extraction from visual imagery, radiomic features, and ordinal aware fusion methods. By creating the new lesion-centric graph-based modules that are expected to follow this research, this system has the

capability to evolve toward being a completely explainable and clinically trusted decision support tool for DR diagnosis.

REFERENCES

- [1] M. Tian, H. Wang, Y. Sun, S. Wu, Q. Tang, and M. Zhang, "Fine-grained attention knowledge-based collaborative network for diabetic retinopathy grading," *Heliyon*, vol. 9, no. e17217, 2023. doi:10.1016/j.heliyon.2023.e17217.
- [2] S. J. Sidiq and T. Benil, "A lightweight transfer learning based ensemble approach for diabetic retinopathy detection," *International Journal of Information Management Data Insights*, vol. 5, p. 100372, 2025. doi:10.1016/j.jjimei.2025.100372.
- [3] J. L. Vázquez Noguera, J. C. Mello-Román, D. P. Pinto-Roa, S. Gómez, D. Pinto-Roa, and others, "Multiclass diabetic retinopathy classification of eye fundus images small datasets performance improvement – a neuroevolution approach," Preprint, 2025. :contentReference[oaicite:0]index=0
- [4] P. K. Das and S. Pumrin, "Diabetic retinopathy classification: Performance evaluation of pre-trained lightweight CNN using imbalance dataset," *Engineering Journal (ENGJ)*, 2024. :contentReference[oaicite:1]index=1
- [5] G. Sharma, V. Anand, R. Chauhan, H. Pokhariya, S. Gupta, and G. Sunil, "Multiclass identification and classification of diabetic retinopathy using transfer learning: A comprehensive approach for enhanced diagnostic," in *Proceedings of the IEEE International Conference on Information Communication Technology (I2CT)*, 2024. doi:10.1109/i2ct61223.2024.10544084 :contentReference[oaicite:2]index=2
- [6] P. D. Poojitha, T. Darak, U. Samaddar, S. Chakradhar Nossam, A. Milak, and S. Palaniswamy, "Multi-Modal detection and classification of diabetic retinopathy," in *2025 International Conference on Electronics, Computing, Communication and Control Technology (ICECCC)*, May 2025. doi:10.1109/ICECCC65144.2025.11064184 :contentReference[oaicite:3]index=3

group3_architecture.png

Fig. 1. Overview of the proposed multimodal DR severity prediction framework. The pipeline integrates CNN feature extractors, radiomics descriptors, cross-attention fusion, and dual-head prediction.